

SECSR2043 OPERATING SYSTEMS

[20 Marks]

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Marks

Instruction: Please answer all of the following questions. Whenever the 🙋 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: *ahmad.sh*).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*. *.; cp ../fruits/*. *.;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]

```
lam@secr2043:~$ nano lam.sh
lam@secr2043:~$ chmod +x lam.sh
lam@secr2043:~$ lam.sh
lam.sh: command not found
lam@secr2043:~$ ./lam.sh
./lam.sh: line 8: cd: drinks: No such file or directory
cp: '../fruits/accord.c' and './accord.c' are the same file
cp: '../fruits/apple.txt' and './apple.txt' are the same file
cp: '../fruits/civic.c' and './civic.c' are the same file
cp: '../fruits/orange.txt' and './orange.txt' are the same file
cp: '../fruits/proton.c' and './proton.c' are the same file
```

```

lam@secre2043:~$ ls -R $HOME
/home/lam:
activity3    fruits      mainprocess.c    share
bye.txt      greeting.txt multiprocessing    subprocess1
cars         hello.txt   multiprocessing.c subprocess1.c
dates        helloworld.jar myfile.txt        subprocess2
drinks       lam.sh      nano.959.save     subprocess2.c
example.txt  mainprocess os_lab

/home/lam/activity3:
a.out echo echo.c hello hello.c sum sum.c

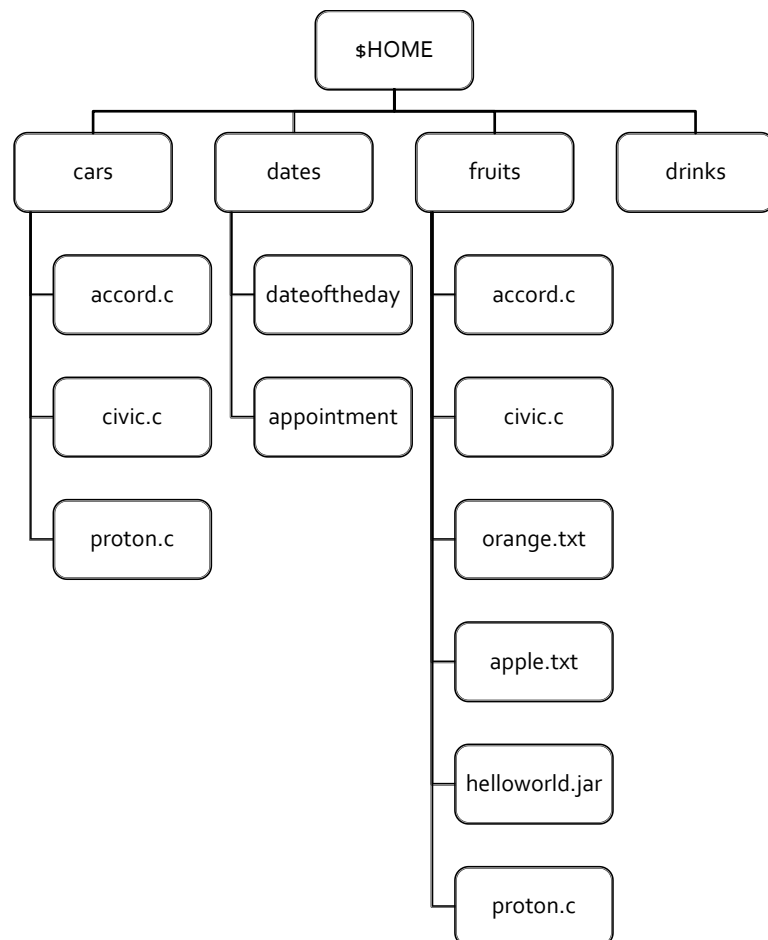
/home/lam/cars:
accord.c civic.c proton.c

/home/lam/dates:
appointment dateoftheday

/home/lam/drinks:

/home/lam/fruits:
accord.c    civic.c    orange.txt
apple.txt  helloworld.jar proton.c

```



- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter “c” as the input to the bash script. [4 marks]

```
#!/bin/bash

echo "Enter the file extension (without the dot, e.g.,
'txt', 'c'): "
read extension

if [ -z "$extension" ]; then
    echo "You must enter a file extension."
    exit 1
fi

files=$(find . -type f -name ".*$extension")

if [ -z "$files" ]; then
    echo "No files found with the .$extension
extension."
else
    echo "$files"

    count=$(echo "$files" | wc -l)
    echo "Total number of .$extension files: $count"
fi
```

```
lam@secr2043:~$ nano readextension.sh
lam@secr2043:~$ chmod +x readextension.sh
lam@secr2043:~$ ./readextension.sh
Enter the file extension (without the dot, e.g., 'txt', 'c'):
c
./activity3/echo.c
./activity3/sum.c
./activity3/hello.c
./fruits/civic.c
./fruits/accord.c
./fruits/proton.c
./multiprocessing.c
./subprocess1.c
./subprocess2.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
./mainprocess.c
Total number of .c files: 13
```



2. The following Figure 1 illustrates a tree structure of some directories and files.

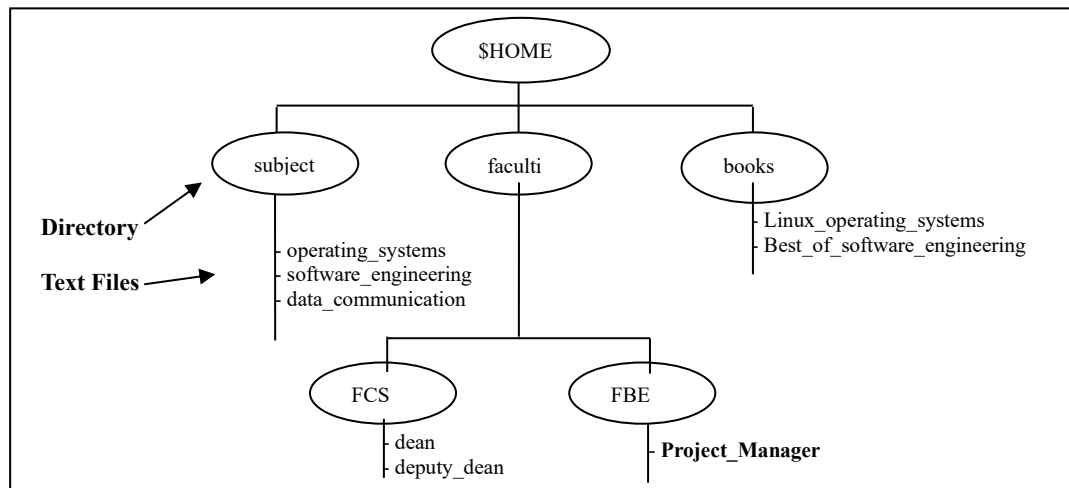


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

```

mkdir subject faculti books;

cd subject;
echo "Operating Systems" > operating_systems;
echo "Software Engineering" > software_engineering;
echo "Data Communication" > data_communication;

cd ../faculti;
mkdir FCS FBE;
cd FCS;
echo "Dean" > dean;
echo "Deputy Dean" > deputy_dean;

cd ../FBE;
echo "Project Manager" > Project_Manager;

cd ../../books;
echo "Linux Operating System" > Linux_operating_systems;
echo "Best of Software Engineering" >
Best_of_software_engineering;

```

```

lam@secr2043:~$ nano myname2a.sh
lam@secr2043:~$ chmod +x myname2a.sh
lam@secr2043:~$ ./myname2a.sh
lam@secr2043:~$ ls -R $HOME
/home/lam:
activity3      hello.txt      os_lab
books          helloworld.jar readextension.sh
bye.txt        lam.sh         share
cars           mainprocess    subject
dates          mainprocess.c  subprocess1
drinks         multiprocessing subprocess1.c
example.txt    multiprocessing subprocess2
faculty        myfile.txt     subprocess2.c
fruits         myname2a.sh
greeting.txt   nano.959.save

/home/lam/books:
Best_of_software_engineering  Linux_operating_systems

/home/lam/faculty:
FBE  FCS

/home/lam/faculty/FBE:
Project_Manager

/home/lam/faculty/FCS:
dean  deputy_dean

/home/lam/subject:
data_communication  operating_systems  software_engineering

```

- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called book.

[2 marks]

Directory/File	Access Control
books	drwxr-xr-x
subjects	drwxr-xr-x
Best_of_software_engineering	-rw-r--r--
FCS	drwxr-xr-x
project_manager	-rw-r--r--

- c) Write another bash script (called myname2c.sh) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

```
#!/bin/bash
```

```
chmod 744 subject
chmod 520 books/Best_of_software_engineering
chmod 607 faculty/FCS
chmod 031 faculty/FBE/Project_Manager
```



```
lam@secr2043:~$ nano myname2c.sh
lam@secr2043:~$ chmod +x myname2c.sh
lam@secr2043:~$ ./myname2c.sh
lam@secr2043:~$ ls -l
drwxr--r-- 2 lam lam 4096 Jun 17 04:02 subject
```

```
lam@secr2043:~$ ls -l books
total 8
-r-x-w---- 1 lam lam 29 Jun 17 04:02 Best_of_software_engineering
-rw-r--r-- 1 lam lam 23 Jun 17 04:02 Linux_operating_systems
lam@secr2043:~$ ls -l faculty
total 8
drwxr-xr-x 2 lam lam 4096 Jun 17 04:02 FBE
drw----rwx 2 lam lam 4096 Jun 17 04:02 FCS
lam@secr2043:~$ ls -l faculty/FBE
total 4
-----wx--x 1 lam lam 16 Jun 17 04:02 Project_Manager
```

- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwxr--r--
Best_of_software_engineering	-r-x-w----
FCS	drw----rwx
project_manager	-----wx--x

End of Lab 3

*** All the Best for Final Exam ***